

The Program in Action

When you run *tictactoe.py*, the output will look like this:

```
Welcome to Tic-Tac-Toe!

  | | 1 2 3
--+-+
  | | 4 5 6
--+-+
  | | 7 8 9
What is X's move? (1-9)
> 1

X| | 1 2 3
--+-+
  | | 4 5 6
--+-+
  | | 7 8 9
What is O's move? (1-9)
--snip--
X|O|X 1 2 3
--+-+
X|O|O 4 5 6
--+-+
O|X|X 7 8 9
The game is a tie!
Thanks for playing!
```

How It Works

To represent tic-tac-toe boards in this program, we use a dictionary with keys '1' through '9' for the spaces on the board. The numbered spaces are arranged in the same way as a phone's keypad. The values in this dictionary are the string 'X' or 'O' for a player's mark and ' ' for an empty space.

```
1. """Tic-Tac-Toe, by Al Sweigart al@inventwithpython.com
2. The classic board game.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: short, board game, game, two-player"""
5.
6. ALL_SPACES = ['1', '2', '3', '4', '5', '6', '7', '8', '9']
7. X, O, BLANK = 'X', 'O', ' ' # Constants for string values.
8.
9.
10. def main():
11.     print('Welcome to Tic-Tac-Toe!')
12.     gameBoard = getBlankBoard() # Create a TTT board dictionary.
13.     currentPlayer, nextPlayer = X, O # X goes first, O goes next.
14.
15.     while True: # Main game loop.
16.         # Display the board on the screen:
17.         print(getBoardStr(gameBoard))
```